WSABI Summer Research Lab	**Summer 2026**

Lab 21: Game Theory

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21.1 Simultaneous Game: Penalty Kicks

A kicker chooses *Left*, *Center*, or *Right*. At the same time, the keeper goes *Left*, *Center*, or *Right*. The kicker's utility is the probability of scoring the penalty. The keeper's utility is the probability of a save, so this is a zero-sum game.

Equal-Strength Penalty Taker

Suppose the scoring probabilities are

	Keeper Left	Keeper Center	Keeper Right
Kick Left	0.60	0.83	0.95
Kick Center	0.74	0.55	0.74
Kick Right	0.95	0.83	0.60

1. List the kicker's pure best response to each keeper action.
2. List the keeper's pure best response to each kicker action.
3. Which actions receive positive probability in equilibrium?
4. Solve for the mixed-strategy Nash equilibrium. At the end, any strategy that receives 0 probability can be eliminated.
5. Compute the kicker's scoring probability at equilibrium.

Imbalanced Penalty Taker

Now suppose the kicker is noticeably stronger to the right:

	Keeper Left	Keeper Center	Keeper Right
Kick Left	0.60	0.83	0.90
Kick Center	0.72	0.54	0.72
Kick Right	0.96	0.89	0.66

6. Again determine which actions receive positive probability in equilibrium.
7. Solve for the new mixed-strategy equilibrium.
8. Compare this equilibrium to the equal-strength case. Which player's mix changes, and why?

21.2 Sequential Game With Complete Information: Bunting

Consider a late-inning situation in which the batter is deciding whether to show bunt. The payoff is the batter's expected runs from the plate appearance plus the resulting base-out state.

The order of moves is:

1. The batter chooses an opening look: *Bunt look* or *Regular look.*
2. The defense responds with *Corners in* or *Normal depth.*
3. If the batter opened with a bunt look, it can then either *Bunt* or *Pull back and swing.* If it opened with a regular look, it can still *Square late and bunt* or *Swing away.*

After a bunt look, the batter's payoffs are:

	Corners in	Normal depth
Bunt	0.18	0.48
Pull back and swing	0.44	0.34

After a regular look, the batter's payoffs are

	Corners in	Normal depth
Square late and bunt	0.12	0.37
Swing away	0.51	0.36

1. After each opening look, identify the batter's best final action against each possible defensive response.
2. Using backward induction, determine the defense's best response after each opening look.
3. Determine the subgame-perfect equilibrium path.
4. Briefly explain why this game is solved differently from the simultaneous penalty-kick game.

21.3 Incomplete Information: Poker on the River

On the river, the bettor can be one of two types:

$$t \in \{\text{Value}, \text{Bluff}\}.$$

The caller does not observe the bettor's type directly.

Suppose the caller begins with prior

$$\mathbb{P}(\text{Value}) = \frac{1}{2}, \qquad \mathbb{P}(\text{Bluff}) = \frac{1}{2}.$$

From population tendencies, the caller treats the following betting frequencies as known:

$$\mathbb{P}(\text{Bet} \mid \text{Value}) = \frac{3}{4}, \qquad \mathbb{P}(\text{Bet} \mid \text{Bluff}) = \frac{1}{4}.$$

If the caller faces a bet, use the following utilities measured from the caller's decision point:

$$u(\text{Fold}) = 0, \qquad u(\text{Call}, \text{Value}) = -100, \qquad u(\text{Call}, \text{Bluff}) = 100.$$

1. Compute the posterior probability $\mathbb{P}(\text{Value} \mid \text{Bet})$.
2. Compute the expected utility of calling after a bet. Should the caller call or fold?
3. Compute $\mathbb{P}(\text{Value} \mid \text{Check})$. What does a check say about the bettor's type?
4. Extension: suppose the pot is 100 before the river bet, a value hand always bets, and a bluff bets with probability b. If calling wins 200 against a bluff and loses 100 against a value hand, solve for the bluff rate b that makes the caller indifferent after observing a bet.